

Limits, L'Hôpital's rule, and epsilon delta definitions

The rigorous foundation under everything above.

THE IDEA

Every derivative in this course was secretly a **limit** — the value a quantity approaches as dx goes to zero. This chapter makes that precise and gives you tools for the awkward cases.

A limit is what a function approaches, not necessarily what it reaches. The function may not even be defined at that point — the limit can still exist.

THE THREE SITUATIONS

Case	What to do
Normal	Substitute the value directly
0/0 that factors	Factor, cancel the common term, then substitute
0/0 that resists	Use L'Hôpital's rule : differentiate top and bottom separately, then substitute

Warning: L'Hôpital's rule differentiates numerator and denominator *independently*. It is not the quotient rule, and it only applies to 0/0 or ∞/∞ .

EPSILON-DELTA, IN ONE SENTENCE

The formal definition says: for any tolerance you demand on the output (epsilon), there is a closeness on the input (delta) that guarantees it. It is the rigorous way of saying "gets arbitrarily close".

WORKED EXAMPLES

CHAPTER 7 — WORKED EXAMPLES

limits and L'Hopital's rule

WHAT A LIMIT IS

the value a function APPROACHES — not necessarily reaches

EXAMPLE 1 simple substitution

$$\lim (x^2 + 3) \text{ as } x \rightarrow 2 = 4 + 3 = 7$$

EXAMPLE 2 the 0/0 case — factor it

$$\lim (x^2 - 4) / (x - 2) \text{ as } x \rightarrow 2$$

substituting gives 0/0 → undefined, so factor :

$$(x - 2)(x + 2) / (x - 2) = x + 2$$

$$\rightarrow 2 + 2 = 4$$

EXAMPLE 3 L'HOPITAL'S RULE

if you get 0/0 , differentiate top and bottom separately :

$$\lim \sin(x) / x \text{ as } x \rightarrow 0 \text{ gives } 0/0$$

$$\text{top } \sin x \rightarrow \cos x$$

$$\text{bottom } x \rightarrow 1$$

$$\rightarrow \cos(0) / 1 = 1$$

note : differentiate them SEPARATELY — this is not the quotient rule.

Why it matters for ML: limits are why derivatives exist at all. They also explain practical failures — division by near-zero, log of near-zero, and the numerical-stability tricks (adding a small epsilon) you will see throughout ML code.

FORMULAS FROM THIS CHAPTER

Limit	the value approached, not necessarily reached
0/0 that factors	factor, cancel, substitute
L'Hôpital	differentiate top and bottom separately
Applies to	0/0 and ∞/∞ only

TEST — Chapter 7

1. Evaluate $\lim (3x + 1) \text{ as } x \rightarrow 4$. $\rightarrow 13$
2. Evaluate $\lim (x^2 - 9)/(x - 3) \text{ as } x \rightarrow 3$. $\rightarrow$ **factor:** $(x-3)(x+3)/(x-3) = x+3 \rightarrow 6$
3. Use L'Hôpital on $\lim (e^x - 1)/x \text{ as } x \rightarrow 0$. $\rightarrow$ **top:** e^x , **bottom:** $1 \rightarrow e^0/1 = 1$